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REVIEW

Impacts and management strategies for crop production in waterlogged or flooded soils: A review

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Abstract

Globally, flooding is one of the most damaging abiotic stresses, besides drought, that affects 17 million km^2 of land surface annually. Recent research indicates that climate change is resulting in more extreme weather events, such as flooding or soil waterlogging, that negatively affect crop production. Therefore, it is imperative to understand how flooding stress affects crops and to develop improved production practices that make cropping systems more resilient and able to cope with extreme weather events. This review paper summarizes the current state of knowledge on the impacts of flooding or soil waterlogging on crop production losses, nitrogen (N) losses, and provides potential management strategies to reduce these losses. The factors affecting the extent of flooding injury in plants as well as plant adaptations under waterlogging stress are also discussed briefly. For the purpose of this review, "flooding" refers to the situation when all or part of the plant is submerged under water, whereas "soil waterlogging" refers to the situation where soil pores are saturated with water. Soil waterlogging also promotes soil N losses through runoff, leaching, and denitrification. Potential management practices that can be used to mitigate soil waterlogging stress include the use of flood-tolerant varieties, adjusting management practices, improving drainage, and practicing adaptive nutrient management strategies. However, these might be site- or crop-specific management practices and they should be validated for their economic viability before developing future management plans that promote sustainable crop yields from waterlogged soils.

1 | INTRODUCTION

Because agronomic crop production is highly sensitive to weather, climate change has affected the agricultural sector globally (Aderonmu, 2015). Climate change has caused modifications in total seasonal precipitation as well as variations within and between seasonal precipitation events. Water availability extremes, including drought and flooding, have increased in agricultural regions of the world over

Abbreviations: BMP, Best Management Practice; CDSI, controlled drainage and subirrigation; EEF, enhanced efficiency fertilizer; ET, evapotranspiration; Fv/Fm, ratio of variable fluorescence to material fluorescence; GIS, Geographic Information System; NBPT, *N*-(*n*-butyl) thiophosphoric triamide; NI, nitrification inhibitor; NUE, nitrogen use efficiency; PCU, polymer-coated urea; UI, urease inhibitors.

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the past 50 years (Aderonmu, 2015; Bailey-Serres, Lee, & Brinton, 2012). The term "flooding" can be used to describe when all or part of a plant is under water (Bailey-Serres et al., 2012), and can be divided into either submergence or waterlogging (Sullivan et al., 2001). Globally, flooding affects over 17 million km^2 of land surface annually, causing severe damage to plants and causes crop production losses (Voesenek & Sasidharan, 2013). An estimated 10–12% of the agricultural area is affected by waterlogging or severe soil drainage constraints globally (Shabala, 2011). The map of a total of 3,713 flood events from 1985 to 2010 recorded in the public global database maintained by G.R. Brakenridge (http://floodobservatory.colorado.edu/) is shown in Figure 1 and maps for recent flood events are available at the Dartmouth Flood Observatory website (http://floodobservatory.colorado.edu/RecentMapping.html). In the United States, flooding is a severe hazard, and was ranked second after drought among abiotic stresses contributing to crop production losses over a 12-yr period from 2000 to 2011 (Bailey-Serres et al., 2012). About 16% of soils are affected by waterlogging in the United States alone (Boyer, 1982).

More than $74 billion (€60 billion) was estimated to be the cost of annual damage from dramatic floods occurring around the world (www.dartmouth.edu/~floods/Archives/2005sum.htm). Within the United States, excessive flooding caused monetary losses of $1.6 billion for corn (*Zea mays* L.) and soybean [*Glycine max* (L.) Merr] production in the Midwestern states in 2011 (Bailey-Serres et al., 2012). During the 1993 floods in the Midwestern region of the United States, crops on over 4 million ha were damaged, and roughly 50% of the total losses from flooding was due to crop damage valued at about $6–8 billion (Rosenzweig, Iglesias, Yang, Epstein, & Chivian, 2001; Rosenzweig, Tubiello, Goldberg, Mills, & Bloomfield, 2002). The NASA simulation models predicted crop production losses worth $3 billion per year by 2030, based on current trends in global climate change characterized by extreme weather events (Rosenzweig et al., 2002). Increases in extreme precipitation events are highly likely in the future as a result of increasing greenhouse gas emissions as assessed by the Intergovernmental Panel on Climate Change (IPCC; Cubasch et al., 2001).

Waterlogging also promotes soil nitrogen (N) loss through processes such as denitrification, nitrate leaching and runoff (Kopyra & Gwó d, 2004) as well as reduces soil N mineralization rates (Haddad et al., 2013; Pengthamkeerati, Motavalli, Kremer, & Anderson, 2006). Nitrogen fertilization is essential in most cropping systems for optimizing crop yields and economic returns (Halvorson, Snyder, Blaylock, & Del Grosso, 2014) as N is generally the most limiting nutrient for crops (Robertson & Vitousek, 2009). However, N losses due to waterlogging reduce N uptake by plants and nutrient use efficiency and ultimately affects surface and groundwater quality. Therefore, it is important to understand plant responses to waterlogging stress and identify the potential management strategies to mitigate the adverse effects of the waterlogging stress. Most previous review articles on soil waterlogging have focused on the plant responses and adaptions to soil waterlogging. Therefore, the primary objective of our paper is to review and summarize the available literature on the causes of soil waterlogging, impacts of flooding or soil waterlogging on production of major crops including corn, soybean, cotton, and wheat (*Triticum aestivum*) in the United States and N losses and provide the potential management strategies that can be implemented to mitigate the crop production losses caused by the waterlogging stress. Limited information is available on the potential management practices to mitigate crop production losses due to waterlogging in the row crops. The factors affecting flooding injury to plants and plant responses and adaptations in response to waterlogging stress are briefly discussed. A general overview and layout of this review paper are shown in Figure 2.

Core Ideas

- Flooding is the most damaging abiotic stress, besides drought, that affects crop production.
- Impacts of flooding on crop production and N losses are reviewed.
- Plant adaptations under waterlogging stress are also discussed.
- Management practices which can be used to mitigate soil waterlogging stress are discussed.

2 | CAUSES OF SOIL WATERLOGGING

Soil waterlogging in the plant-rooting zone is influenced by several factors including climate, factors that affect the amount of water entering into the soil, the amount of water moving through or over the surface of the soil, and use of soil water by plants and other organisms. A major reason for the increase in flooding or waterlogging is the increase in the frequency of extreme precipitation events and multiple wet days (Kunkel, 2003). For example, a major flooding of the Mississippi River in the United States in 1993 was an outcome of saturated soils and elevated stream flows created by extreme precipitation in June and July along with a period of abnormal precipitation during the previous seven months from November to May (Kunkel, 2003). This flood also contributed to the doubling of the Gulf's "Dead Zone" in 1993 due to increased nutrient runoff into the Mississippi River and Gulf of Mexico (Rosenzweig et al., 2001). Precipitation amount, number of

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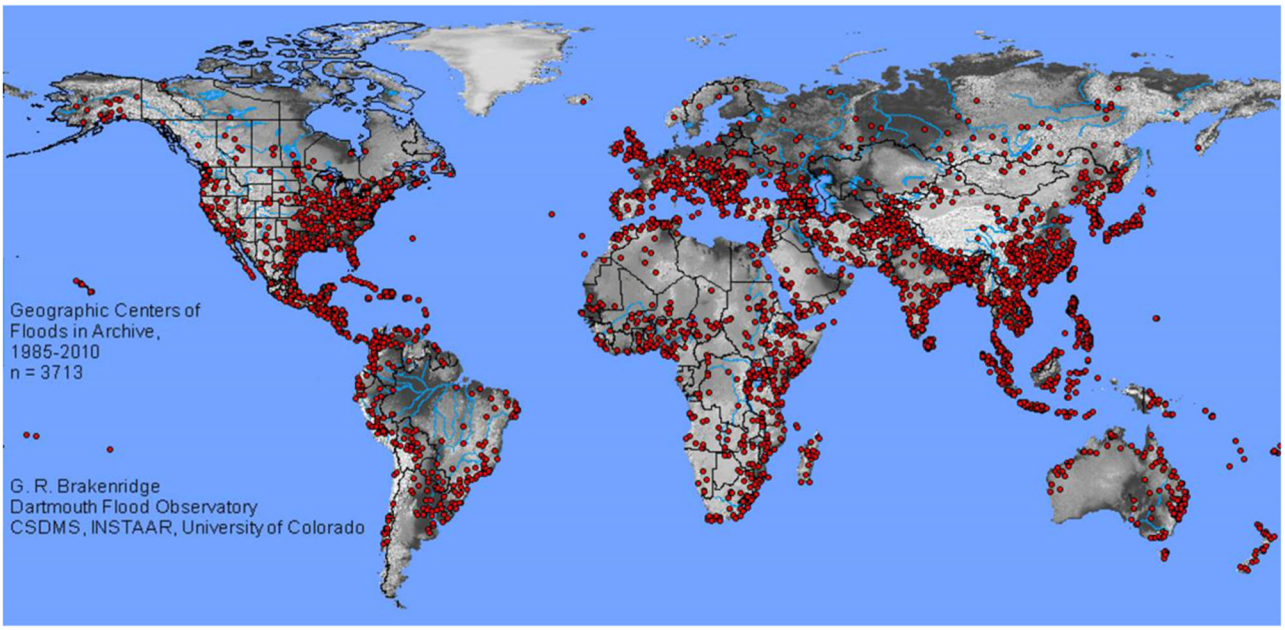


FIGURE 1 The map of total 3,713 flood events from 1985 to 2010 recorded in the public global database maintained by G.R. Brakenridge (http://floodobservatory.colorado.edu/)

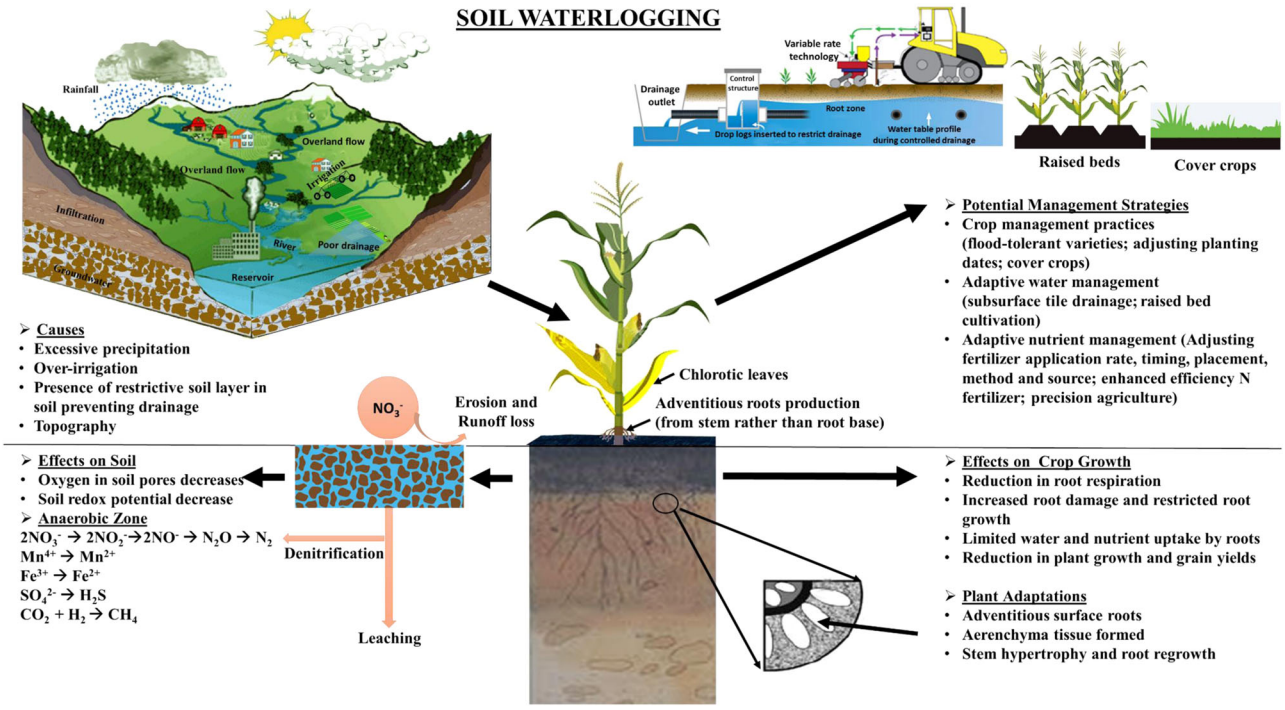


FIGURE 2 Overview of soil waterlogging causes, crop production and nitrogen losses, and potential management strategies

precipitation events, event size, and time between events differentiates extreme wet years from dry years (Knapp et al., 2015). In the upper Midwestern United States, the frequency of days with heavy precipitation has increased by 50% during the 20th century (Groisman, Knight, & Karl, 2001). During extreme wet years, precipitation amounts differed from average years by approximately 40% in regions receiving 1,000-mm mean annual precipitation (Knapp et al., 2015); with multiple extreme daily precipitation events occurring as often as twice compared to average or dry years. Human alterations in land use and stream channels, in addition to increased precipitation frequency and intensity,

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are another reason for increases in flooding (Kunkel, 2003).

Other causes of soil waterlogging or flooding include over-irrigation, rainfall occurring after irrigation (Kirkpatrick, Rothrock, Rupe, & Gbur, 2006), land topography (Hanna, Harlan, & Lewis, 1982), shallow water table, soil compaction (Batey, 2009), poor surface and internal drainage in soils such as in heavy clays soils, claypan soil, or duplex soils (coarse-textured topsoil over compacted clay subsoil; Kirkpatrick et al., 2006; Scott, DeAngulo, Daniels, & Wood, 1989; Shaw, Meyer, McNeill, & Tyerman, 2013). Increased runoff from upslope topographic positions can cause waterlogging in low-lying positions, particularly when the low-lying area has poorly drained soils (Hanna et al., 1982, Singh, Williard, & Schoonover, 2016). Poor soil structure caused by natural factors or human activity, such as soil compaction from heavy traffic or puddling, can reduce downward movement of water resulting in shallow perched water tables within the top few centimeters of soil or in the subsoil causing waterlogging (Batey, 2009). Soil compaction caused by tractor wheel traffic in the field can decrease infiltration, permeability, and water flow through the soil profile and can cause flooding. Soil compaction also increases impedance to root growth and can influence the seed germination, emergence, and growth of crops. Compaction causes a rearrangement of soil particles, changes bulk density, aggregate stability and/or arrangement, and affects soil structure, which subsequently changes the movement of air in the soil profile. Soil preparation for rice (*Oryza sativa*) cultivation to create flooding conditions for rice in Asian countries results in subsoil compaction, which causes poor drainage and consequently, soil waterlogging problems for the succeeding wheat crop (Samad, Meisner, Saifuzzaman, & van Ginkel, 2001).

Heavy clay soils with swelling-shrinking clay types are susceptible to soil waterlogging in events of high precipitation or irrigation, for example: Vertisols in the Mississippi River Delta Region. These soils mostly contain montmorillonitic clay which are characterized by relatively high cation exchange capacity (Blessitt, 2007; Han, Kingery, Hargreaves, & Walker, 2007; Pettry, Switzer, 1996). Sharkey soil, the dominant soil series mapped in the Mississippi River Delta, occupies ~400,000 ha in the state of Mississippi and >1.2 million ha in the United States and are very deep, poorly or very poorly drained soils (USDA-NRCS 2006). Heavy clay soils having high water holding capacity and poor internal drainage can impede infiltration into the soil profile due to swelling when soil reaches maximum water holding capacity (Blessitt, 2007).

Flooding can also be caused by restriction of free drainage in the soil profile due to the presence of natural restrictive soil layers, such as claypan subsoil layers, that reduce infiltration. Worldwide, an area of approximately 290 million hectares has soils with restrictive clay subsoil horizons (USDA-NRCS, 2006). Hollinger (1995) reported that most farmland soils in the Midwestern United States are poorly drained (Figure 3A). For example, in the Midwestern United States, the total area under poorly-drained claypan soils is estimated to be around 4.0 million ha, with principle extensive areas in northern Missouri, southeastern Kansas, and southern Illinois and less extensive areas in Nebraska, Oklahoma, and southern Iowa where soils also have subsoil restrictions similar to those of claypan soils (Jamison, Smith, & Thornton, 1967; Smith & Whitt, 1948). Out of this total claypan soil area, 3.3 million ha of poorly-drained claypan soils are in Missouri and Illinois, known as the central claypan areas (MLRA 113; USDA-NRCS, 2006), which may have waterlogging in the topsoil layers after rainfall events (Anderson, Gantzer, & Brown, 1990; Figure 3; Jamison et al., 1967). In claypan soils, there is usually an abrupt, 100% increase in clay content in the subsoil horizon that occurs within a small vertical distance in the soil profile compared to the overlying soil horizons (Motavalli, Anderson, & Pengthamkeerati, 2003). Depending upon the landscape position, depth to the claypan layer can vary from 10 cm at the backslope position to 40 cm at the footslope positions (Jiang, Anderson, Kitchen, Sadler, & Sudduth, 2007). The clay content can be greater than 460 g kg^{-1} in the argillic horizon in the claypan soils (Jamison & Peters, 1967). Presence of this subsurface argillic horizon can alter drainage due to lower permeability, and can result in a perched water table and soil waterlogging in the upper soil layers during extreme precipitation events in spring (Motavalli et al., 2003). In addition, a perched water table results in short-term waterlogging in sandy duplex soils in Australia where rainfall infiltrates rapidly through a sandy topsoil but accumulates above a compacted clay subsoil which has a low hydraulic conductivity at 5- to >100-cm depth (Tennant, Scholz, Dixon, & Purdie, 1992).

3 | CROP PRODUCTION LOSSES DUE TO SOIL WATERLOGGING

3.1 | Crop yield losses

Drought and flooding accounted for more than 70% of the crop yield reductions in the United States in 2011 (Bailey-Serres et al., 2012; Milly, Wetherald, Dunne, & Delworth, 2002). The rising trend in precipitation has increased crop damage and off-farm loss of soil and nutrients from excess water (Morton, Hobbs, Arbuckle, & Loy, 2015). Soil waterlogging resulting from extreme precipitation events is another major contributor to the crop production losses, in addition to floods (Kozdrój & van Elsas, 2000) (Figures 4a and 4b). Crop production losses due to temporarily flooded or saturated soils are a persistent problem in the Midwestern United States (Luce, 2015; Wiebold, 2015). In upland areas,

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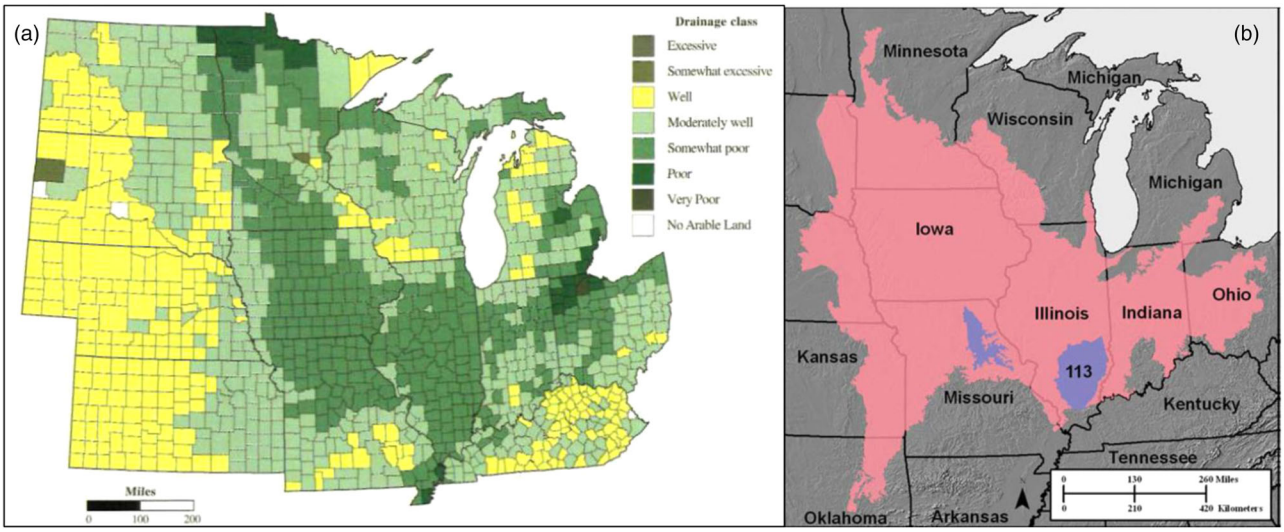


FIGURE 3 Drainage classification of farmland in the Midwestern states (Hollinger, 1995) (a) and the central claypan area (blue color) in Missouri and Illinois (USDA-NRCS, 2006) (b)



FIGURE 4 Photos of flooded corn fields in Missouri, (a) flooded soybean fields in Mississippi, (b) mud deposition on soybean plants after flooding, (c) and complete plant loss in low-lying flooded areas in corn plots (d)

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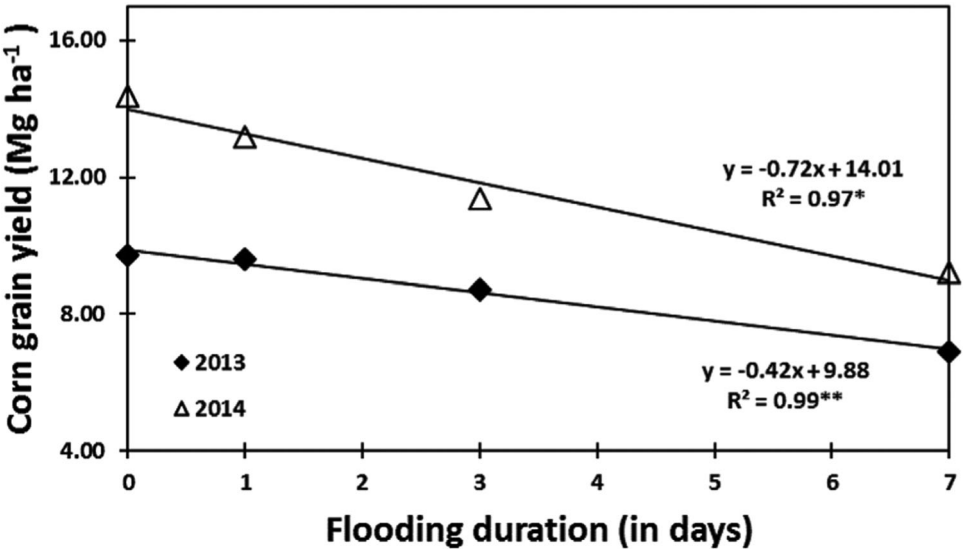


FIGURE 5 Decrease in corn grain yield due to flooding duration in poorly drained claypan soils in northeastern Missouri (Kaur et al., 2017). * Significant at the .05 probability level; ** Significant at the .01 probability level

excessive rainfall created saturated soil conditions that caused about 70% of the crop losses during the 1993 Mississippi floods (Rosenzweig et al., 2002). In poorly-drained claypan soil, each day of waterlogging at a V6 (corn plant with six leaves) growth stage (Abendroth, Elmore, Boyer, & Marlay, 2011) of corn reduced grain yields by 0.57 Mg ha^{-1} d^{-1} when data were averaged over 2 yr (Kaur, Zurweller, Nelson, Motavalli, & Dudenhoeffer, 2017) (Figure 5). Soybean grain yields were reduced by 20–39% due to soil waterlogging for 8 d at the R5 (pod fill reproductive stage; Fehr & Caviness, 1977) growth stage (Rhine, Stevens, Shannon, Wrather, & Sleper, 2010). A decrease in corn and soybean yields was also reported by Singh et al. (2016) due to soil waterlogging in low-lying topographic positions (Singh et al., 2016). Soil waterlogging at vegetative and reproductive stages resulted in soybean seed yield reduction of 17–43% (Oosterhuis, Scott, Hampton, & Wullschleger, 1990) and 50–56% (Scott, DeAngulo, Wood, & Pitts, 1990), respectively. Urban, Roberts, Schlenker, and Lobell (2015) reported that soil waterlogging during the spring season caused 1–3% yield loss for corn and soybean crops in three U.S. states including Illinois, Iowa, and Indiana, over the 62-yr study period from 1950 to 2011. The percent yield reductions due to waterlogging or flooding stress at different growth stages in corn, soybean, wheat, and cotton (*Gossypium hirsutum*) are summarized in Table 1. Based upon previous literature, yield losses ranging from 1 to 100% can occur due to waterlogging stress, depending upon the crop, waterlogging duration, and crop growth stage (Table 1). However, the impact of soil waterlogging on crop yields can vary in different regions due to the variation in regional climatic and economic conditions, soil types, plant species and age, and management differences.

Other effects of flooding that affect crop growth and ultimately, yields include deposition of mud and crop residues on the plants (Figure 4c), sand deposition, and soil crusting. Foliage suffers physically from strong flow rates or abrasion due to suspended particles (Jackson & Ram, 2003). Flooding stress causes plants to wilt within a few hours to 4 d when there is high transpiration (Kozlowski, 1984). Excessive soil moisture leads to increased risk of plant disease and insect infestation (Urban et al., 2015), such as crazy top or downy mildew (*Sclerophthora macrospora*), gray leaf spot (*Cercospora zeae-maydis*), and common smut (*Ustilago maydis*) in corn; scab in wheat (*Fusarium graminearum*); and sudden death syndrome (*Fusarium virguliforme*) in soybeans. Another issue with soil waterlogging is the inability to operate machinery in wet conditions which can delay planting or harvesting (Urban et al., 2015), and consequently can reduce crop yields. For example, delayed planting due to excessive rainfall in April and May or replanting of corn as late as mid-June decreased corn yields by 12–22% between planting times in late April or early May and early to mid-June in Ohio (Lindsey, Thomison, Mullen, & Geyer, 2015).

3.2 | Physiological and metabolic changes in plants due to waterlogging

3.2.1 | Respiration

Soil waterlogging inhibits plant growth by creating hypoxic (oxygen deficient) or anoxic (complete absence of oxygen) conditions in the soil. An overview of changes in soil due to excessive soil moisture is reported in Figure 2. The diffusion rate of oxygen in water is approximately 1/10,000th that of air,

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TABLE 1 Yield losses due to waterlogging stress at different growth stages of major row crops of the United States

Crop	Timing of waterlogging initiation[a]	Waterlogging duration d	Yield loss[b] %	References
Corn	V6	1, 3, 7	10–36	Kaur et al. (2017)
	V3	7	16–38	Kaur (2016)
	V6, VT	1 ,2, 3	1–22	Singh & Ghildyal (1980)
	V2, V7, VT, R1	10	8–80	Zaidi et al. (2004)
	20 and 40 d after planting	0, 11	51–69	Sandhu, Singh, Singh, & Khera (1986)
	V6, VT	5	25–43	Shah et al. (2012)
	V1, V6, VT, R1	2, 4, 6, 8, 10	9–100	Liu et al. (2013)
	V3,V6, 10 d after VT	3, 6	7–33	Ren et al. (2014)
	V3	-	40	Bhan (1977)
	Early and late vegetative, flowering	10	19–64	Mukhtar, Baker, & Kanwar (1990)
	4 weeks after planting	1, 2, 3, 4, 5, 6	6–61	Chaudhary, Bhatnagar, & Prihar (1975)
	6- and 30-inch height of corn and at silking	1, 2, 3	2–33	Ritter & Beer (1969)
	V6	3, 6, 9, 12	27–56	Ahmad (1991)
Soybean	V5, R1, R5	2, 4, 6, 8	20–41	Rhine et al. (2010)
	V4, R2	2, 4, 7, 14	20–84	Scott et al. (1989)
	V4, R2	7	39–52	Oosterhuis et al. (1990)
	V1, V4, R2	7	12–56	Scott et al. (1989)
	V2, V3, V7, R1, R3, R5, R6, R6.3	7	9–93	Linkemer et al. (1998)
	V2, V3	3, 6, 8	20–93	Sullivan et al. (2001)
Wheat	3–4 leaf stage	10, 20, 30	44	Collaku & Harrison (2002)
	Leaf 7 to anthesis	5	34–92	Romina, Abeledo, & Miralles (2014)
	2 wk after planting and ear emergence	42	40–53	Watson, Lapins, & Barron (1976)
	Flowering	5, 10, 15, 20, 25, 50	4–90	Olgun, Metin Kumlay, Cemal Adiguzel, & Caglar (2008)
	Anthesis and Grain Ripening	10, 20, 30	40	Ghobadi & Ghobadi (2010)
	Seeding growth and stem elongation	10	8–39	Yavas, Unay, & Aydin (2012)
	Stem elongation	4, 8, 12, 16, 20, 24	25–50	Marti, Savin, & Slafer (2015)
	Grain filling	20, 30	15–23	Luxmoore, Fischer, & Stolzy (1973)
	25 d after planting	1, 2, 4, 6	8–39	Sharma & Swarup (1989a)
	Late tillering stage	14, 28	16–49	Arguello et al. (2016)
	3 and 4 leaf stage	4, 8, 12, 16, 20, 40, 60	19–30	Pampana, Masoni, & Arduini (2016)
Cotton	Seedling, squaring, flowering, and boll opening	2, 4, 6, 8, 10	7–39	Wang et al. (2017)
	Flowering, boll development	8, 16	13	Hocking, Reicosky, & Meyer (1985)

[a]Growth stages description for the corn and soybean plants can be obtained from Abendroth et al. (2011) and Fehr and Caviness (1977).

[b]Yield loss is compared to non-flooded control.

and the flux of O_2 into soils is approximately 320,000-times less in water-filled soil pores compared to gas-filled soil pores (Armstrong & Drew, 2002; Colmer & Flowers, 2008). In plants, respiration is the main metabolic process affected by soil waterlogging. Oxygen is an electron acceptor in the oxidative phosphorylation process generating energy as adenosine triphosphate (ATP) and regenerating NAD^+ from NADH, which is used as a reducing power for biochemical pathways (glycolysis; Dennis et al., 2000). The NAD^+ and NADH are the oxidized and reduced forms of nicotinamide-adenine dinucleotide (NAD). In absence of oxygen, plants undergo fermentation process for ATP production as a rescue mechanism (Drew, 1997), which reduces ATP production efficiency as fermentation produces only three ATP molecules compared

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TABLE 2 Corn and soybean responses to soil waterlogging stress at different growth stages

Crop	Growth Stages[a]	Plant responses to waterlogging stress	References
Corn	V1	Reduced plant height and leaf area	Liu et al. (2013)
	V2	High plant mortality; reduced plant height, leaf area, dry weight, stem and total carbohydrate content, short anthesis–silking interval	Zaidi et al. (2004)
		Reduced N, P, K, and Zn concentrations in leaves; ears per plant; and grain weight per ear	Sandhu et al. (1986)
	V3	Reduced leaf greenness and stomatal conductance	Kaur (2016)
		Decreased grains per ear, 1000-grain weight, plant height, ear height, leaf area, grain-filling rate, and dry matter accumulation	Ren et al. (2014)
	V6	Chlorosis	Kaur et al. (2017)
		Retarded growth, reduced plant height, N and K uptake, increased P uptake	Singh & Ghildyal (1980)
		Decrease in total leaf number, leaf area, and dry matter accumulation	Shah et al. (2012)
		Reduced plant height and leaf area	Liu et al. (2013)
		Decreased grains per ear, 1000-grain weight, plant height, ear height, leaf area, grain-filling rate, and dry matter accumulation	Ren et al. (2014)
		Reduced leaf area, root length, and weight	Ahmad (1991)
	V7	High plant mortality; reduced plant height, leaf area, dry weight, stem and total carbohydrate content; short anthesis–silking interval	Zaidi et al. (2004)
	VT	Retarded growth, plant height, N and K uptake, increased P uptake	Singh and Ghildyal (1980)
		High plant mortality and short anthesis–silking interval	Zaidi et al. (2004)
		Decrease in total leaf number, leaf area, and dry matter accumulation	Shah et al. (2012)
		Reduced plant height and leaf area	Liu et al. (2013)
	R1	Reduced total carbohydrate content	Zaidi et al. (2004)
		Reduced plant height and leaf area	Liu et al. (2013)
Soybean	V1	Reduced N and K concentrations	Scott et al. (1989)
		Reduced plant population and plant height	Sullivan et al. (2001)
	V2	Reduced plant height and branch number	Linkemer et al. (1998)
		Reduced plant population and plant height	Sullivan et al. (2001)
	V3	Reduced plant height	Linkemer et al. (1998)
	V4	Reduced N and K concentrations and canopy height, reduced dry matter accumulation, yellowing, and abscission of leaves, stunting	Scott et al. (1989)
		Reduced photosynthesis, dry matter accumulation, growth and stomatal closure	Oosterhuis et al. (1990)
	V5	Reduced plant growth and biomass accumulation	Cho and Yamakawa (2006)
	V6	Reduced height, chlorosis, stunting; diminished plant growth	Griffin & Saxton (1988)
	V7	Reduced branch number	Linkemer et al. (1998)
	R1	Suppressed nodule nitrogenase and leaf nitrate reductase activities	Sung (1993)
		Reduced branch number and reduced pods per branch	Linkemer et al. (1998)
		Reduced shoot dry weight and leaf area	Youn et al. (2008)
	R2	Reduced N and K concentrations and canopy height; higher Mn, Fe, Al concentrations	Scott et al. (1989)
		Reduced canopy height and dry matter accumulation	Scott et al. (1989)
		Reduced photosynthesis, dry matter accumulation, growth, and stomatal closure	Oosterhuis et al. (1990)
	R3	Reduced seed size and reduced pods per branch	Linkemer et al. (1998)
	R5	Reduced seed size	Linkemer et al. (1998)
		Suppressed nodule nitrogenase and leaf nitrate reductase activities	Sung (1993)

[a]Growth stages description for the corn and soybean plants can be obtained from Abendroth et al. (2011) and Fehr and Caviness (1977).

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to 39 ATP molecules through respiration of hexose molecule (Geigenberger, 2003). Sharp reduction in ATP formation by fermentation process impairs cellular metabolism and function (Geigenberger, 2003). Fermentation results in lowered cytosolic pH, induction of glycolysis, and accumulation of lactate and ethanol (Drew, 1997). Under flooded soil conditions, plants follow three fermentation pathways depending upon cytosolic pH including ethanol, lactic acid, and a plant-specific pathway involving alanine aminotransferases to produce alanine from glutamate and pyruvate. Lack of oxygen in waterlogged soils for root respiration reduces root growth. Soil phytotoxins also inhibit root formation and promote root decay in flooded soils. Due to the waterlogged conditions, roots are not able to exploit a large soil volume and tend to grow near the soil surface (Sairam, Kumutha, Ezhilmathi, Deshmukh, & Srivastava, 2008). Ren, Zhang, Dong, Liu, and Zhao (2016) reported that root length, root length density, and the number of root tips in corn decreased with 6 d of waterlogging at the V3 (corn plant with three leaves), V6, and 10 d after VT (tasseling) growth stages when compared to the nonwaterlogged control. Flooding stress on 2-d-old soybean plants significantly suppressed fresh root weight, whereas glycolysis, fermentation, stress, and cell wall-related protein synthesis were increased (Oh & Komatsu, 2015). Flooding also reduces nodulation and nitrogenase activity in soybean due to a reduction in oxygen supply (Henshaw, Gilbert, Scholberg, & Sinclair, 2007; Maekawa, Shimamura, & Shimada, 2011). A list of plant responses and injury symptoms to waterlogging stress in corn and soybean are provided in Table 2. Under waterlogged conditions, water and nutrient uptake is reduced due to lower root conductance ultimately resulting in poor plant growth from nutrient deficiency (Steffens, Hütsch, Eschholz, Lošák, & Schubert, 2005).

Flooding stress also alters soluble sugar concentrations in the plants (Gupta & Kaur, 2005). Sugars, such as sucrose and glucose, act as substrates needed for cellular respiration and as osmolytes for maintaining cell homeostasis, whereas fructose is responsible for the synthesis of secondary metabolites and erythrose-4-P, which is used in the synthesis of lignin and phenolic compounds (Gupta & Kaur, 2005). Mutava et al. (2015) reported that flooding stress significantly increased glucose concentration in the leaves, but decreased sucrose and raffinose concentration in a flood-susceptible soybean cultivar. Alteration in sugar status due to abiotic stresses in plants affects gene expression by down- and up-regulating their expression.

3.2.2 | Photosynthesis

The waterlogging injury also reduces photosynthesis (Huang, Bridges, Nesmith, & Johnson, 1994; Mutava et al., 2015; Oosterhuis et al., 1990), possibly due to stomatal closure, abscisic acid (ABA), ethylene, and active oxygen species production (Jackson, Young, & Hall, 1988). Jackson and Ram (2003) observed that photosynthesis was CO_2 limited in waterlogged plants. The partial stomatal closure was a response to waterlogging to prevent leaf water deficits and wilting, rather than being a response to low leaf water potential. The stomatal closure also limits CO_2 in plant cells and consequently, results in accumulation of oxygen free radicals. Plants suffer oxidative stress under waterlogged conditions through an increase in reactive oxygen species, such as superoxide (O_2^-), singlet oxygen (1/2 O_2), hydrogen peroxide (H_2O_2), and hydroxyl radicals (OH^-) produced in chloroplasts during the electron transport along the electron chain (Jackson & Colmer, 2005; Subbaiah & Sachs, 2003; Zheng et al., 2009). The reactive oxygen species are usually present in plants at various levels under normal metabolism and can contribute significantly to waterlogging damage in plants. The highly reactive oxygen species alter normal cellular metabolism through oxidative damage to lipids, proteins, and nucleic acids and can further damage Photosystem II under severe stress conditions (Alscher, Donahue, & Cramer, 1997; Imlay, 2003).

Waterlogging damages photosynthetic apparatus which resulted in a lower ratio of variable to maximal fluorescence (Fv/Fm), and reduced photosynthetic rate and transpiration, especially at the tillering and jointing-boot growth stages in wheat (Shao et al., 2013). The ratio of Fv/Fm is a good measure of photo-inhibitory impairment of plants under environmental stresses, including waterlogging. A reduction of Fv/Fm showed that there is either reversible photoprotective downregulation or an irreversible inactivation of photosystem II (PSII) during waterlogging. Flooding stress reduced chlorophyll content in soybeans by 18–34% (Mutava et al., 2015). The leaf area index of corn was decreased by an increase in waterlogging duration (Liu, Liu, Xiao, Nan, & Gong, 2013). Waterlogging at the post-anthesis stage significantly reduced wheat root respiratory activity, leaf greenness (SPAD reading), photosynthetic rate, stomatal conductance, transpiration rate, grain number per spike, 1,000-grain weight, grain yield, and increased intercellular CO_2 concentration (Wu et al., 2014).

4 | FACTORS AFFECTING THE EXTENT OF WATERLOGGING INJURY

4.1 | Timing of flooding during the life cycle of the crop

Soil waterlogging can reduce plant emergence (Fausey & McDonald, 1985). Singh and Ghildyal (1980) reported that soil flooding of corn at V6 was more harmful than at the

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VT growth stage. Corn in later vegetative or reproductive growth stages (V7, VT, R1) were less susceptible to soil waterlogging damage compared to early growth stages (V2) when the growing point was still below or close to the soil surface (Ren et al., 2014; Ren et al., 2016; Ritter & Beer, 1969; Zaidi, Rafique, Rai, Singh, & Srinivasan, 2004). Ren et al. (2016) found that corn grain yields decreased 26, 21, and 13% due to a 6-d waterlogging period at V3, V6, and 10 d after the VT stage, respectively. This indicated that the V3 stage was more susceptible to waterlogging damage compared to V6 and VT stages. Shah, Srivastava, da Silva, and Shahi (2012) also found higher yield losses in corn due to waterlogging at V6 (43%) compared to the VT (25%) growth stage. In soybean, grain yield reductions were greater when soil was waterlogged at R1 or R5 compared to the V5 growth stage (Rhine et al., 2010). Another study showed that the early vegetative (V2) and reproductive stages (R1, R3, R5) of soybean were most affected by the soil waterlogging compared to the late vegetative (V3, V7) and reproductive (R6) growth stages (Linkemer, Board, & Musgrave, 1998). These research results indicate that crop growth stage at the time of flooding or waterlogging is an important determinant of the magnitude of waterlogging stress injury to the plants.

4.2 | Frequency and duration of flooding

Crop production losses become more severe with an increase in flood duration (Kaur et al., 2017; Musgrave, 1994). An increase in soil waterlogging duration from 1 to 7 d at the V6 growth stage of corn caused 0.57 Mg ha^{-1} reduction in grain yield with each day of waterlogging (Kaur et al., 2017; Figures 5 and 6). Three and 7 d of waterlogging at the V6 growth stage of corn resulted in 10–21% and 29–36% reduction in yields in two years of this study (Kaur et al., 2017). An increase in waterlogging duration from 4 to 8 d reduced the final plant biomass by an additional 10%, indicating the adaptive response of corn to waterlogging for survival. Wheat grain yields were reduced by 20 and 24% when waterlogged for 44 and 58 d at 93 and 64 d after wheat sowing, respectively (Dickin & Wright, 2008). The leaf area index of corn was decreased by an increase in waterlogging duration (Liu et al., 2013).

4.3 | Soil and water temperatures during flooding

Crop damages due to flooding increased with a rise in temperature from 10 to 25°C (Fausey & McDonald, 1985). In that study, plant survival was affected by the interaction between flooding duration and temperature. Nielsen (2015) observed that corn plants at the V6 growth stage survived for 2–4 d under flooding at 15°C, but plants died at temperatures above 25°C. Generally, plants that survive flooding will show new leaf growth in about 3–5 d following favorable temperatures. Wu et al. (2013) observed that high temperature accelerated the negative effects of waterlogging on wheat at post-anthesis. The negative effects of waterlogging and temperature on wheat growth were ranked: waterlogging + high temperature > waterlogging > high temperature. Similarly, Trought and Drew (1982) found that the waterlogging damage to wheat plants of the same chronological age was greater at high soil temperature (10–18°C). The damage to plants at the same growth stage was similar if the plants were exposed to high temperatures over a short period of waterlogging compared to when they are subjected to low soil temperature for a longer waterlogging period (Trought & Drew, 1982). Flooding injury to soybean seed increased with increasing flooding duration from 1 to 48 h during germination at 25°C compared to no increase in injury at 15°C (Wuebker, Mullen, & Koehler, 2001).

5 | PLANT ADAPTATIONS TO WATERLOGGING STRESS

Plants have several adaptive mechanisms to cope with waterlogging or flooding stress, such as the formation of airspace (aerenchyma) in the root cortex, stem enlargement (hypertrophy), radial root oxygen barrier, adventitious root formation (especially near the soil surface), and early root tip death (Bailey-Serres et al., 2012; Subbaiah & Sachs, 2003; Zaidi et al., 2004). Aerenchyma is the term used for plant tissues containing enlarged gas spaces to provide a low-resistance internal pathway for gas exchange between aerobic shoots and anaerobic roots (Evans, 2004). Further details about aerenchyma formation are provided in Evans (2004). Aerenchyma helps the plant in delivering oxygen to the root tip and rhizosphere as well as remove gases (CO_2, ethylene, and methane) from the root and soil (Colmer & Flowers, 2008; Shannon, White, Lawson, & Gilmour, 1996) and prevent the influx of toxic compounds including nitrites and sulfides of iron, copper, and manganese by rhizosphere oxygenation (Armstrong & Armstrong, 1988). Aerenchyma formation causes the death of some cortex cells which reduces oxygen demand. Many studies have reported aerenchyma formation in response to soil waterlogging in multiple crops (Campbell, Sojka, & Karlen, 1984). In corn roots, aerenchyma formation is mainly initiated by ethylene produced either endogenously or applied exogenously (Drew, 1997; Jackson, 1985).

Adventitious roots are formed from the submerged part of the stem in flooded plants and grow horizontally close to the surface so that oxygen is more available to these roots compared to belowground roots (Sairam et al., 2008). Diffusion of gases between shoots and roots is facilitated by the presence

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FIGURE 6 Effects of flooding duration on corn growth on poorly drained claypan soil in northeast Missouri. Corn was flooded at V6 (corn plant with 6 leaves) growth stage. (Kaur et al., 2017)

of large airspaces in the adventitious roots. Adventitious roots often contain more aerenchyma compared to the primary lateral roots and thus are less affected by the anaerobic conditions caused by waterlogging (Visser, Nabben, Blom, & Voesenek, 1997). Details about adventitious root formation are provided by many researchers (Bragina, Rodionova, & Grinieva, 2003; Sairam et al., 2008; Visser et al., 1997). Studies have also shown an increased contribution of the apoplastic bypass to water flow and reduced dependence of roots on the hypoxia-sensitive aquaporin-mediated water transport in the adventitious roots (Calvo-Polanco, Señorans, & Zwiazek, 2012).

The complex defensive antioxidant system in plants can protect cellular membranes and organelles from oxidative damaging effects of reactive oxygen species (Foyer, Lelandais, Galap, & Kunert, 1991). Antioxidant enzymes, including superoxide dismutase, catalase, and various peroxidases, such as guaiacol peroxidase, ascorbate peroxidase, and glutathione reductase can effectively minimize the activity of reactive oxygen species (Foyer et al., 1991). In addition to antioxidant enzymes, other antioxidant compounds, such as ascorbate, glutathione, and β-carotene also play an important role in the removal of toxic oxygen compounds (Burke & Mahan, 1991; Hodges, Andrews, Johnson, & Hamilton, 1996).

6 | EFFECTS OF WATERLOGGING ON SOIL AND NITROGEN AVAILABILITY

6.1 | Changes in soil due to waterlogging

Oxygen is the most important electron acceptor for microbial decomposition of organic C compounds in soil and sediments due to its ample supply in the atmosphere, easy diffusion into soil pores and ease of reduction (Patrick & Jugsujinda, 1992). Under anoxic conditions, the reduction of O_2 is followed by NO_3^-, Mn^{4+} and Fe^{3+}, SO_2^- and CO_2. Microbial activity mainly controls redox reactions in soils (Kögel-Knabner et al., 2010). Reducing conditions in the soil can be quantified by measuring redox potential (E_h). Redox potential measurements also help to determine whether chemical compounds, such as NO_3^-, are present in their oxidized form or in chemically reduced forms which may affect N losses such as N_2O emissions resulting from denitrification (Vepraskas & Cox, 2002). Soil redox potential normally varies between −300 and +900 mV, whereas E_h of reduced soils can be between −300 mV to +400 mV (Husson, 2013; Sahrawat, 2005). Redox potentials for oxic (aerobic), suboxic, and anoxic (anaerobic) soil conditions are >414, 414–120, and <120 mV, respectively, at soil pH 7 (Essington,

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2004). Fageria, Carvalho, Santos, Ferreira, and Knupp (2011) reported that reduction of NO_3^- to dinitrogen gas takes place at a soil E_h of 220–280 mV. Soil E_h varies depending upon multiple factors including the amount of organic matter, the nature and content of electron acceptors, temperature, pH, and the duration of submergence (Ponnamperuma, 1972). For example, Kaur (2016) reported that soil E_h decreased by 50 mV with each day of waterlogging in poorly-drained claypan soils, indicating the development of oxygen reduction and anaerobic conditions in the soil. However, a study by Unger, Motavalli, and Muzika (2009) showed the development of anaerobic soil conditions within 12 d of waterlogging in 5-wk flooding treatments as indicated by a decrease in redox potential and dissolved oxygen (Unger et al., 2009).

Soil pH also changes along with the redox potential in flooded soils (Kögel-Knabner et al., 2010). Soil pH increases due to consumption of protons in acidic soils, whereas pH decreases in alkaline soils due to the accumulation of carbon dioxide after flooding, which neutralizes alkalinity (Sahrawat, 2005). It may take days to weeks for changes in soil pH to occur due to flooding, which depends on the soil type, source and amount of organic matter, soil microbial population, temperature, and other soil chemical properties (Kongchum, 2005). Soil redox potential change of 59 mV is accompanied by a one-unit change in pH according to the Nernst equation (Kögel-Knabner et al., 2010). In a study conducted by Kaur (2016), soil surface pH increased from 5.6 to 6.1 after 3 d of waterlogging, then decreased to 5.9 on the 7th d of waterlogging at the end of the flooding treatment in only one of the 3 yr of the research study. However, Unger et al. (2009) found no significant effect of flooding on soil pH. Soil temperature also changes due to flooding. Temperature controls rates of reactions. Few studies have shown that soil temperature of waterlogged treatments were higher than the nonwaterlogged treatments (Kaur, 2016; Unger et al., 2009). In contrast, Zurweller, Motavalli, Nelson, and Dudenhoeffer (2015) found no differences in soil temperature at a 10-cm depth between nonwaterlogged and 3-d waterlogged soils. These changes in redox potential, soil pH, and temperature occurring due to waterlogging ultimately affects N transformation and availability in the soil.

6.2 | Nitrogen losses from soil

Soil waterlogging causes reduction in nitrate concentrations in soil (Unger et al., 2009; Zurweller et al., 2015). For instance, Unger et al. (2009) found a reduction in NO_3–N concentrations following 3–5 wk of soil waterlogging. Zurweller et al. (2015) reported a rapid decline in soil NO_3–N concentration with each day of waterlogging amounting to a 61% lower soil NO_3–N concentration in waterlogged treatments than in a nonwaterlogged control. The reduction in soil NO_3–N concentrations resulting from waterlogging might occur due to N losses by various processes discussed below.

6.2.1 | Runoff losses

Nitrogen losses in agricultural fields may occur through surface runoff and erosion processes. Previous studies have reported <5% of total N applied to be lost through surface runoff in U.S. Midwestern claypan soils (Blevins, Wilkison, Kelly, & Silva, 1996; Ghidey & Alberts, 1999). In a 7-yr (1991–1997) study on claypan soils with corn–soybean rotations, one year with precipitation of 142% of normal precipitation accounted for 44 and 46% of total N and NO_3–N losses, respectively, in runoff out of total losses observed during the 7-yr study period (Udawatta, Motavalli, Garrett, & Krstansky, 2006). This indicates an increase in soil runoff losses with an increase in precipitation amount and intensity.

6.2.2 | Denitrification

Denitrification is the process of reduction of NO_3^- to molecular N or other gaseous forms of N by both heterotrophic and autotrophic bacteria, known as 'denitrifiers.'

$$2NO_3^- \rightarrow 2NO_2^- \rightarrow 2NO^- \rightarrow N_2O \rightarrow N_2$$

Denitrification process occurs under anaerobic soil conditions caused by waterlogging or flooding. Soil surface N_2O emissions show positive correlations with soil temperature, water-filled pore space, and mineral N content in soil and are increased by soil waterlogging (Allen, Kingston, Rennenberg, Dalal, & Schmidt, 2010).

Soils with restrictive layers to drainage, such as claypan soils, are more susceptible to N losses via denitrification due to the formation of a perched water table in the surface soil layers because of the low hydraulic conductivity of the claypan subsoil (Motavalli et al., 2003). Aerobic N transformations in claypan soils are limited by the wet conditions due to combinations of poor drainage and excessive rainfall events, especially in spring season (Pengthamkeerati et al., 2006). In poorly drained claypan soils, a study conducted by Zurweller et al. (2015) showed that a major proportion of cumulative surface soil N_2O emissions occurred mostly during and shortly after a soil flooding event. In the same research, the proportion of N fertilizer lost as N_2O–N varied from 0.03 to 0.04% in nonwaterlogged soils and 1.1–2.6% in the waterlogged soils during the 2-yr field experiment. However, Pomes, Wilkinson, and McMahon (1998) reported that low rates of denitrification in the claypan soils resulted in leaching of excess N remaining in the cropped soil for months or years.

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6.2.3 | Nitrate leaching

Nitrate leaching is the process of movement of NO_3–N through the rooting zone. Due to its high solubility and relative lack of surface retention in the soil, NO_3–N is highly susceptible to both lateral and vertical transport by water through the soil profile. Around 10–40% of applied N can be lost through leaching depending upon the application time, fertilizer source, and the crop being fertilized (O' Legg & Meisinger, 1982). Restricted water movement due to the lower permeability of argillic horizons reduces NO_3–N leaching in soils with subsoil restrictions such as claypan soils (Jung, Kitchen, Sudduth, & Anderson, 2006). Preferential flow paths developed by biological (e.g., worm burrowing or root growth) or physical processes (e.g., desiccation) are other pathways for rapid movement of N to shallow groundwater (Puckett, 1995; Wilkison & Blevins, 1999). Flooding and waterlogging can result in greater NO_3–N leaching in low cation exchange capacity and sandy texture soils where residual or surplus N concentration are higher after harvesting the cash crop (Salazar, Vargas, Nájera, Seguel, & Casanova, 2014). Huber, Luster, Bernasconi, Shrestha, and Graf Pannatier (2012) reported that during flooding events when water fluxes are high, nitrate from soils can contribute up to 11% to the total nitrate load in groundwater.

6.3 | Effects of waterlogging on N uptake and assimilation

Reduced root conductance due to waterlogging can lower N absorption and consequently, limit N transport and distribution in the plants, with a reduction in the concentrations of N, P, K, Mg, and Zn in leaves and stems and increased concentrations of these elements in the plant root system (Huang et al., 1994). Multiple studies have reported N deficiency symptoms in plants under waterlogging stress. Board (2008) reported that soybean yield was reduced under soil waterlogging mainly due to N deficiency. Negative correlation of flood duration with soybean leaf N concentration was reported by Sullivan et al. (2001). Bacanamwo and Purcell (1999) observed that N fixation in soybean was more sensitive to waterlogging than biomass accumulation, as a reduction in N fixation occurred faster than the decrease in biomass accumulation. Transient N deficiency occurred in wheat due to waterlogging stress, even when the plants were supplied with N at the time of planting (Robertson, Zhang, Palta, Colmer, & Turner, 2009). Reduction in N uptake, N recovery, and agronomic efficiency due to waterlogging was also observed in cotton and Bahiagrass (*Paspalum notatum*; Hodgson & MacLeod, 1988; Sigua, Williams, Chase Jr, Grabowski, & Kongchum, 2013). The decrease in the greenness (17–27 SPAD units) of corn leaves (as measured with chlorophyll leaf meter) with an increase in flooding duration from 1 to 7 d at the V6 growth stage of corn (Figure 6) was observed by Kaur et al. (2017). Chlorophyll meter leaf readings were shown to be strongly correlated with leaf chlorophyll concentration or greenness and with leaf N status (Markwell, Osterman, & Mitchell, 1995; Scharf, Brouder, & Hoeft, 2006).

6.4 | Use of N fertilization for recovery from waterlogging injury

Plant tolerance to waterlogging stress may increase by N fertilizer applications as they may enhance and accelerate plant adaptive mechanisms to waterlogging, such as adventitious root growth and root regrowth after a flooding event. Under waterlogged conditions, N losses through denitrification and leaching may cause N deficiencies, decrease N availability, and inhibit root function resulting from low oxygen levels under waterlogged conditions that may reduce plant N uptake (Nielsen, 2015).

Beneficial effects of N application for crop recovery from waterlogging stress have been observed in many crops. For example, corn yields decreased by 20% due to 72 h of waterlogging in plots treated with high N fertilizer rate (312 kg N ha^{-1}) compared to 35% reduction in plots treated with low N fertilizer (45 kg N ha^{-1}; Ritter & Beer, 1969). In Missouri, the beneficial effect of a post-waterlogging N fertilizer application (at V10 growth stage; 84 kg N ha^{-1}) was found after up to 3 d of waterlogging initiated at the V6 corn growth stage (Kaur et al., 2017). Waterlogging stress in corn genotypes was reduced by the use of sodium nitroprusside as a donor of NO in flooding water, with higher response in a waterlogging susceptible genotype (Jaiswal & Srivastava, 2015). In contrast, Ashraf (1999) found that supplementary nitrate in the growth medium of 42-d-old corn plants under long-term waterlogging for 21 d in a greenhouse study had an injurious effect on growth. In a greenhouse study, nitrate addition every 3 d when plants were 18-d old reduced flood sensitivity of soybean plants compared to unfertilized soybean (Bacanamwo & Purcell, 1999; Oliveira, Freschi, & Sodek, 2013). Wu et al. (2014) concluded that adverse effects caused by 6 d of waterlogging stress in winter wheat at 3–9 d after anthesis stage can be alleviated by supplementary N applications at 5 d after anthesis growth stage (Wu et al., 2014). Supplementary foliar N applications improved wheat root respiratory activity, leaf greenness, photosynthetic rate, stomatal conductance, transpiration rate, 1,000-grain weight, grain yields, protein content, and increased the activities of enzymes such as nitrate reductase (NR), glutamine synthetase in flag leaf and grain, and glutamic-pyruvic transaminase in grain (Wu et al., 2013; Wu et al., 2014). Supplemental N ameliorates root respiration and photosynthetic rate in wheat by promoting synthesis of enzymes for photosynthesis,

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inhibiting abscisic acid production and stimulating gibberellic and cytokinin synthesis (Zongbin, Jianguo, & Xiaochun, 2008). Greater N rates applied prior to waterlogging promoted root vigor of cotton seedlings after waterlogging stress in a growth chamber study (Zhou & Oosterhuis, 2012). Cotton yields were increased by pre-irrigation foliar N application under severe waterlogging conditions when soil N levels were not high (Hodgson & Chan, 1982; Hodgson & MacLeod, 1988).

7 | POTENTIAL MANAGEMENT PRACTICES FOR REDUCTION OF CROP PRODUCTION AND N LOSSES DUE TO WATERLOGGING

Management practices that prevent crop production and/or N losses under waterlogged soil conditions may include adjusting management practices (e.g., early or late planting; use of flood-tolerant crop varieties, and cover crops); adaptive nutrient management practices (e.g., enhanced efficiency fertilizers [EEFs], rescue N applications); altering the application rate, timing, and placement (Dawar, Zaman, Rowarth, Blennerhassett, & Tumbull, 2011a); use of adaptive water management practices (e.g., improving drainage, raised bed planting); or use of precision agriculture technology. These practices may be used alone or in combination to synergistically limit N losses and maximize yields.

7.1 | Crop management practices

7.1.1 | Flood-tolerant crop varieties

One of the approaches for reducing crop production losses is identification, development, and use of crop varieties tolerant to soil waterlogging. Multiple studies have reported genetic variation in many crops for their response to soil waterlogging or flooding (Collaku & Harrison, 2002; Davies & Hillman, 1988; Huang et al., 1994; Torbert, Hoeft, Vanden Heuvel, Mulvaney, & Hollinger, 1993; Zaidi et al., 2004; Zaidi, Maniselvan, Srivastava, Yadav, & Singh, 2010). Flood-tolerance mechanisms of plants may include the ability to transport oxygen to anaerobic roots, high capacity for anaerobic respiration and maintaining glycolytic metabolism, their ability to keep stomata open for photosynthesis, and the ability to avoid oxidative damage (Caudle & Maricle, 2012). Flood-tolerant plants have traits to develop adaptive mechanisms, such as aerenchyma formation, development of adventitious roots, and stem hypertrophy. Traditional breeding programs use wild relatives of cultivated plants as a source of biotic and abiotic stress resistance to improve plant tolerance to stresses. For example, Mano and Omori (2007) used 'teosinte' as a germplasm source for breeding flood-tolerant corn as it can form adventitious roots at the soil surface during extended periods of saturated soils, and can form constitutive aerenchyma in root cortex under nonflooded conditions and inducible aerenchyma under flooded conditions (Mano & Omori, 2007). Additional traits for flood-tolerant crop varieties that may be considered in breeding programs include an increase in plant N uptake, N use efficiency, and improvement in seed or grain quality parameters. Flooding can occur at any growth stage of the crop depending upon the location and weather conditions. Waterlogging caused by rainfall events in fall season can delay crop harvesting and therefore, it is important to breed crop varieties for characteristics such as lower susceptibility to diseases or pests, good seed quality, and stem strength. Flooding in early season also exposes plants to cold soil temperature and therefore, it is also necessary to develop varieties with tolerance to both cold and waterlogging stress. In summary, it is important to develop and test new crop varieties that are tolerant or resistant to multiple biotic or abiotic stresses including waterlogging stress, heat and drought stress, as well as disease pathogens.

7.1.2 | Adjusting planting dates

Planting dates can be adjusted to avoid waterlogging conditions during the early spring season to facilitate good crop emergence and growth, particularly in the Midwestern United States where early spring season precipitation is common. Planting is usually delayed due to the cool and wet soils that are common to this region. Cool, wet soils can delay crop emergence and early plant vigor. Later planting dates may avoid early season extreme precipitation events and waterlogged soil conditions, but it may lead to a shorter growing season and exposure to drought later in the growing season that may also result in the lower crop yields. However, Kucharik (2006) reported that in-season droughts during summer have shifted planting dates earlier into the spring season. Other factors responsible for early planting dates include the development of varieties tolerant to suboptimal temperatures and diseases, seed treatments, improved planting equipment, a crop protection products, and adoption of time-saving crop management practices such as conservation tillage (Kucharik, 2006). Early planting dates allow for a longer growing season resulting in higher yields through an increase in the time duration for absorption of solar radiation and biomass accumulation (Kucharik, 2006). However, there is a risk of crop damage due to suboptimal soil temperature at early planting dates; therefore, crop varieties used for early planting should also be tolerant to low soil temperature that occur during or after planting. Shifting planting dates for soil waterlogging abatement depends upon the timing and duration of waterlogging duration.

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7.1.3 | Cover crops

Long-term use of cover crops may not only improve soil health, but also may decrease waterlogging by improving soil structure, reducing compaction, and increasing the water infiltration rate (Blanco-Canqui et al., 2015). Root channels from cover crops can increase macropores resulting in greater water movement through soil. For example, a cover crop mixture of bromegrass (*Bromus inermis* Leyss.) and strawberry clover (*Trifolium fragiferum* L.) reduced surface soil strength by 38–41% and increased steady infiltration rate and cumulative water intake by 37–41% and 20–101%, respectively, compared to the no-cover crop treatments (Folorunso, Rolston, Prichard, & Loui, 1992).

Winter cover crops can affect water availability for summer crops by decreasing evaporation while using stored water in the soil for transpiration (Smith, Frye, & Varco, 1987). Greater transpiration rate of cover crops during spring can potentially dry soil for early crop planting. Higher water demand of cover crops along with higher temperature during spring can help in removing excess water from the waterlogged soils through greater evapotranspiration (ET) losses.

Multiple studies have reported the effects of cover crops on soil moisture content reduction (Monteiro & Lopes, 2007; Zhang & Schilling, 2006; Zhu, Gantzer, Anderson, Beuselinck, & Alberts, 1991). A 3-yr vineyard study found that volumetric water content under cover crop mixture (60% nonlegume and 40% legume) was lower compared to no cover crops (Monteiro & Lopes, 2007). Zhang and Schilling (2006) studied the effect of land cover on the water table, soil moisture, and groundwater recharge, and reported that reed canary grass (*Phalaris arundinacea*) had a lower water table and soil moisture content through higher ET losses which reduced groundwater recharge (Zhang & Schilling, 2006). In the Coastal Plain region of North Carolina, crimson clover (*Trifolium incarnatum* L.) reduced soil water content 28–55% before corn planting in the top 15-cm soil depth (Ewing, Wagger, & Denton, 1991). Zhu et al. (1991) reported that cover crops had the potential to reduce soil water content if they were allowed to bear seed until late April on poorly drained Mexico silt loam soils of central Missouri. Use of cover crops during winter fallow period can also be a potential option for managing soil waterlogging. However, the effect of cover crops on the distribution of water in soil profile can be positive, negative, or neutral, depending upon the soil type, climatic region, and cover crop species used (Blanco-Canqui et al., 2015). Therefore, use of multiple species of cover crops in areas susceptible to flooding or in areas susceptible to drought within a field should be further evaluated. Agricultural fields do not always have zero grade and can have topography which can influence water and nutrient dynamics within a field and add spatial variability to cover crop biomass production and cash crop yields. Therefore, it is important to further evaluate interaction of topograpghy with cover crops for reducing waterlogging stress in large agricultural fields. The geographic information system (GIS) spatial modeling and remote sensing might be usful for identifying areas susceptible to waterlogging within an agricultural field. Since cover crops generally do not provide direct economic benefits, further research might be needed on the effects of implementing cover crops only on susceptible areas instead of the whole field.

7.2 | Adaptive water management

7.2.1 | Subsurface tile drainage

Drainage can remove excess water from the soil, which improves plant root growth, plant emergence, plant stands, workability of the soil, and consequently, results in higher crop yields (Blevins et al., 1996; Nelson, Meinhardt, & Smoot, 2012; Nelson, Paniagua, & Motavalli, 2009; Nelson, Smoot, & Meinhardt, 2011a). Subsurface drainage technology is used worldwide for improving drainage of waterlogged soils (Nelson et al., 2012; Sharma, Kaledhonkar, Thimmappa, & Chaudhari, 2016). Use of subsurface tile-drain systems with drainage water recycling in the poorly drained soils of the U.S. Midwest is a management option for preventing soil waterlogging that induces N losses through runoff, erosion, or denitrification.

Several studies have examined the interactions of drainage systems with N management (Nelson et al., 2009) and have reported increased crop yields, N uptake, and nitrogen use efficiency (NUE). In the Midwestern United States, free drainage or managed tile drainage systems that use a water table control structure have improved crop yields and reduced fertilizer loss through drainage water flow (Nash, Nelson, & Motavalli, 2015a; Nash, Nelson, Motavalli, & Anderson, 2015b). For example, on poorly drained claypan soils, use of integrated subsurface tile-drained systems at 6.1- and 12.2-m spacing between tile lines increased yields by 11–22% compared to nondrained plots (Nelson, 2017). Another study reported that use of tile-drainage improved soil moisture conditions and allowed planting of soybean earlier by up to 17 d as well as increased soybean yields by 9–22% than nondrained control plots (Nelson et al., 2011a).

Based on a review on impact of agricultural practices on N management and nitrate loss into the water resources, Dinnes et al. (2002) pointed out that artificial drainage systems negatively affect water quality since they act as shallow direct channels to water bodies from agricultural fields. Factors affecting nitrate loss from the tile-drainage agricultural fields include precipitation amount and timing, initial soil moisture content, time of the year, tile depth, and tile spacing (Drury et al., 2009). Subsurface drainage systems

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that incorporate both controlled drainage and subirrigation (CDSI) can improve crop production, N use efficiency, and reduce possible negative effects on the water quality through reduction of nitrate-N entering into surface or groundwater systems (Drury et al., 2009; Fausey, Brown, Belcher, & Kanwar, 1995; Frankenberger et al., 2006). Use of CDSI allows irrigation of root-zone during in-season drought periods, which may help increase crop growth and nutrient uptake, thereby reducing N losses in the tile-drainage water and up to a 75% reduction in nitrate-N loading (Frankenberger et al., 2006). Also, nitrate losses through drainage can be reduced by adopting management practices such as improved N application timing, application rates and techniques, use of enhanced efficiency N fertilizers and soil testing for N application, reducing tillage, utilizing crop rotations, and cover crops (Dinnes et al., 2002). Linking subsurface tile-drainage to on farm irrigation reservoirs is also a promising option to address the water quality issues while simultaneously providing waterlogging and drought resilience to crop production systems. However, this needs to be evaluated further.

7.2.2 | Raised beds

Raised beds are used to deal with water management issues, such as waterlogging or irrigation application in semi-arid and arid regions (Govaerts, Sayre, Lichter, Dendooven, & Deckers, 2007). Raised seedbeds can increase crop yields by maintaining a favorable soil moisture content through improved drainage and provide a conduit for irrigation water application (Velmurugan, Swarnam, Ambast, & Kumar, 2016). Raised beds improve soil physical properties via higher macroporosity and infiltration, lower bulk density, and higher aggregate stability by improving soil structure (Hassan, Hussain, & Akbar, 2005; Roth, Fischer, Piggin, & Meyer, 2005). Restricting traffic in the furrows helps to avoid soil compaction in the beds that further allows better soil infiltration, root growth, and surface and subsurface drainage.

Several studies have reported the beneficial effects of raised bed planting for increasing crop yields under waterlogged soil conditions as compared to flat seedbed planting (Bakker, Hamilton, Houlbrooke, & Spann, 2005; Blessitt, 2007; Siler et al., 2002). Raised beds lowered the potential for waterlogging by increased runoff rates due to the presence of furrows, and improved soil structure as indicated by lower bulk densities and infiltration rates on poorly drained duplex clay soils (Bakker et al., 2005). Raised bed planting reduced waterlogging stress because the top 15 cm of raised bed soils remained unsaturated.

In the United States, raised beds have been evaluated from southern regions to northern areas such as soils of the Red River Valley in in North Dakota, which are highly susceptible to waterlogging upon excessive rainfall events. Hoppe (2013) found that soybean yields in the Red River Valley were not decreased when soybean was planted on the raised seedbed compared to flat seedbeds during a dry year (Hoppe, 2013). In the U.S. Midwest, raised bed planting of corn promoted growth and increased biomass and yields, probably due to a suitable environment for root growth and gas exchange in poorly drained silt loam soil (Siler et al., 2002). In the Mississippi Delta region of the United States, raised bed planting increased plant height, leaf area index by 52–107%, soybean yield by 14–31%, and resulted into 23–45% higher net returns above input costs when compared to soybeans planted on flat seedbeds (Blessitt, 2007). Blessitt (2007) concluded that raised seedbeds can be a feasible option for reducing waterlogging damage on early planted soybeans on poorly drained clay soils in the Mississippi River Delta region of the United States.

In summary, raised beds can be used for preventing waterlogging stress in agricultural areas where tillage is used for land preparation, but it may not be a potential option under no-till cropping systems. However, the use of raised beds along with other management practices such as tile-drainage is not yet evaluated and can be researched to assess combined effect of these two practices on alleviation of waterlogging stress in areas of high waterlogging risk.

7.3 | Adaptive nutrient management

Fertilizer application rate, timing, placement, and sources affects N uptake, losses, and NUE. Synchronization of the N availability with crop N demand and uptake is important for optimum N management to get maximum benefit from the applied fertilizers through increased crop yields while reducing N losses. Selection of optimum N source, application rate, and timing is an important management option for N management in poorly drained soils. A single N application may result in N losses if soil waterlogging occurs from early season extreme precipitation events after pre-plant N application. Corn may need additional N fertilizer applications (known as rescue N) when the preplant-applied N fertilizer is suspected to be lost due to the excessive soil moisture from extreme precipitation events or when wet soil prevented preplant or side-dress N applications (Nelson, Scharf, Stevens, & Burdick, 2011b). In the Midwestern United States, corn yields were highly responsive to N fertilizer applications as late as the silking stage, although full yield potential was not achieved for N application at the silking stage (Scharf, Wiebold, & Lory, 2002). Only about 3% corn yield loss was observed when N applications were delayed until V12–V16 (Scharf et al., 2002).

Corn yield response to rescue N application sources and application methods depends upon many factors, such as leaf injury and gaseous N losses from surface-applied N. Nelson

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et al. (2011b) reported that broadcast application of urea ammonium nitrate or ammonium nitrate on corn at V5–V6, V7–V8 (corn plant with 7–8 leaves) or V9–V10 (corn plant with 9–10 leaves) growth stages reduced corn yields due to the leaf injury caused by broadcast application. They concluded that N source and placement needs to be considered while making decisions for rescue N application when corn is greater than 30-cm tall. Post-flood rescue N application of urea with urease inhibitor *N*-(*n*-butyl) thiophosphoric triamide (NBPT) effectively increased the yield of corn that was waterlogged up to 3 d (Kaur et al., 2017). Therefore, the N management in the areas prone to soil waterlogging should consider split N application methods to avoid N losses. However, the final decision about the N application timing and methods can be made only after considering higher net returns from available options for N application methods or timing. In addition, it is also important to evaluate the impact of waterlogging duration and timing in response to N applications and sources on the extent of N leaching and denitrification losses under different soil textures.

7.3.1 | Enhanced efficiency fertilizers

Increased interest in the use of enhanced efficiency fertilizers (EEFs) has occurred in recent years due to several factors such as increased cost of fertilizers, higher crop value, greater environmental concerns associated with N application, inefficient crop N uptake, and improved manufacturing technologies (Stewart, 2008). An increased interest in EEFs has grown due to unpredictable precipitation events and more extreme precipitation events that may result in greater fertilizer losses. Enhanced efficiency fertilizers are fertilizer products that increase plant uptake and decrease the potential of nutrient losses to the environment (e.g., gaseous losses, leaching, and/or runoff) when compared to a conventional fertilizer source (Motavalli et al., 2008). They can include slow- and controlled-release fertilizers, nitrification inhibitors (NI), and urease inhibitors (UI). Benefits of EEFs include their simplicity and ease of use, availability of different types with diverse characteristics, and their potential for N loss reduction (Motavalli, Goyne, & Udawatta, 2008). A detailed description of EEFs was reviewed by Chen et al. (2008), Motavalli et al. (2008), Shaviv and Mikkelsen (1993), Trenkel (2010), and Timilsena et al. (2015).

Slow or controlled release fertilizers either delay initial nutrient availability or extend nutrient availability for a longer period during the life cycle of a plant by regulating N release in the soil, controlling the availability of ammonium to nitrifying bacteria, and/or reducing nitrate loss through leaching or gaseous losses (Motavalli et al., 2008). The mechanisms responsible for delaying N availability include controlled water solubility of the material by semi-permeable coatings, occlusion, protein materials, or other chemical forms such as slow hydrolysis of water-soluble low molecular weight compounds (Halvorson et al., 2014). Urea-formaldehyde based fertilizers, sulfur-coated urea, and polymer-coated urea (PCU) are some examples of slow- or controlled-release fertilizers. The term "controlled-release" has been generally accepted to refer to coated or encapsulated fertilizers for which the factors determining the rate, pattern, and duration of release are known and regulated during the fabrication, and the term "slow-release" is generally used for microbial decomposed N products such as urea–formaldehyde based fertilizer (Chien, Prochnow, & Cantarella, 2009; Shaviv, 2001).

Nitrification inhibitors are the chemicals that can either slow, delay, or restrict the nitrification process by affecting the metabolism of *Nitrosomonas* spp. bacteria involved in the nitrification process. Nitrification inhibitors slow nitrate production from ammonium, thereby decreasing leaching of nitrate and denitrification losses from soil. The NIs can inhibit biological oxidation of ammonium N to nitrate N from 4 to 10 wk (Halvorson et al., 2014). Some commercially available NIs are nitrapyrin (2-chloro-6-trichloromethyl-pyridine), DCD (dicyandiamide), and DMPP (3,4-dimethylpyrazole phosphate; Huber, Warren, Nelson, & Tsai, 1977), and pronitridine (Schwab, Schaefer, Nelson, Ferguson, & Helmers, 2017).

Urease inhibitors (UI) reduce the urea hydrolyzation rate and its conversion to ammonium by restricting the activity of urease enzyme in the soil. By delaying this hydrolysis, volatility losses of ammonia, which occur primarily at the soil surface can be lowered by delaying urea hydrolysis process through the use of UIs (Motavalli et al., 2008). However, reduction of urease activity by use of UIs may also cause phytotoxicity by an accumulation of excess urea in plant parts (Krogmeier, McCarty, & Bremner, 1989). Globally, more than 14,000 compounds or mixtures have been tested and patented as UIs (Chien et al., 2009). However, NBPT is the most commercially available UI with relatively high efficiency in inhibiting urease at low concentrations in a wide variety of soils compared to other UIs (Kawakami, Oosterhuis, Snider, & Mozaffari, 2012; Rawluk, Grant, & Racz, 2001). In soil, NBPT converts to its oxon analog *N*-(*n*-butyl) phosphoric triamide (NBPTO), which is the actual inhibitor of urease activity (Creason, Schmitt, Douglass, & Hendrickson, 1990; Krogmeier et al., 1989). These chemical forms, NBPT and NBPTO, compete with urea molecules for the enzyme Ni receptor sites in the structure of the urease molecule and inhibits its activity (Kolodziej, 1994). Delaying urea hydrolysis by NBPT allows more time for rainfall or irrigation to move the urea from the soil surface into sub-soil layers near the plant rooting zone, where it is less susceptible to ammonia volatilization losses (Dawar, Zaman, Rowarth, Blennerhassett, & Turnbull, 2011b).

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Many studies have reported that use of EEFs resulted in greater crop yields and lower N losses. Corn yields increased 1–1.2 Mg ha^{-1} by pre-plant application of anhydrous ammonia with nitrapyrin and PCU compared to noncoated urea (Nash, Nelson, & Motavalli, 2013). In eastern Canada, PCU and urea with NI resulted in greater corn yield during wet years compared to urea applied without NI, but showed no differences between yields among fertilizer sources during dry years (Gagnon, Ziadi, & Grant, 2012). Economic analysis revealed that, despite 30% higher costs, PCU gave comparable net returns at an equivalent N rate than urea ammonium nitrate (UAN) in wet years. Anhydrous ammonia and PCU applications increased corn yields by 1.5–1.8 Mg ha^{-1} and gross profits \$50–642 ha^{-1} compared to urea in the low-lying field positions compared to the summit and side slope positions in a poorly-drained claypan soil landscape (Noellsch, Motavalli, Nelson, & Kitchen, 2009). The PCU-treated soil had 51–63% lower soil NO_3–N concentration than urea 59 d after application in water samples collected by suction lysimeters and thus, reduced nitrate leaching early in the season (Nelson et al., 2009). In the United States, the NBPT evaluation from 78 trials showed that it increased corn grain yields by 270 kg ha^{-1} when applied with urea and by 100 kg ha^{-1} when applied with UAN when averaged over N and NBPT rates for all locations and years of the study (Hendrickson, 1992). Similarly, Dawar, Zaman, Rowarth, Blennerhassett, and Turnbull (2010) reported that use of NBPT with urea increased ryegrass herbage dry matter yield and N uptake more than urea alone or other forms of N fertilizers used in the study through improved N availability. Further research should be conducted to evaluate the use of EEFs under varying waterlogging durations, either under continuous or intermittent waterlogging conditions, as well as assessing their N losses under such conditions.

7.4 | Precision agriculture

Different soil moisture regimes may exist within a field due to the spatial variability caused by slope, topography, or soil heterogeneity. Within the same field, low-lying areas may become waterlogged under excessive rainfall events while the upland areas may suffer from droughts in the dry years. For example, Singh et al. (2016) reported that topography resulted in excess soil moisture at the depositional position and reduced corn and soybean yields in southern Illinois. Similarly, Adler et al. (2019) reported that corn, soybean, and cover crop biomass yields were consistently low in the channel and footslope slopes of terraced fields in Missouri due to waterlogged conditions prevailing in the channel and footslopes. Furthermore, N availability and losses also vary with the soil moisture content, which may result in variable crop responses to N fertilizer applications across agricultural landscapes. Anaerobic conditions due to waterlogging reduces soil solution nitrate-N at footslope positions possibly due to N loss through denitrification, which further contributed to lower corn and soybean yields at footslope land scape positions (Singh, 2018; Singh, Williard, Schoonover, Nelson, & Kaur, 2019). Precision agriculture technology can be utilized for identifying management zones based on yield maps, soil productivity maps, high-resolution digital elevation models from LIDAR (Light Detection and Ranging), flow accumulation, and soil electrical conductivity maps for site-specific management to increase crop productivity (King et al., 2005).

The pairing of flood-tolerant crop varieties with or without optimum N fertilizer rates and/or sources with the low productivity areas within a field that are susceptible to occasional waterlogging can be a potential management option for reducing yield and N losses due to land constraints, and needs to be further researched. Also, waterlogging in the low productive zone can be managed by introduction of tile drainage in the targeted areas and should be supplemented with control structures to manage water table depth in the field in order to reduce N loss and increase crop production. Variable rate N fertilizer application practice for improving N recovery, increasing crop yields, and reducing environmental N losses has been used extensively using information from grid sampling, site-specific management zones based on previous yield maps, or optical sensors. Noellsch et al. (2009) reported that PCU resulted in higher corn yields compared to conventional urea in low-lying areas due to lower N loss from PCU during saturated soil conditions in the early spring. Therefore, a variable-source N-fertilizer strategy for targeted application of EEFs and conventional N fertilizer can increase crop production and N recovery efficiency and is a feasible management option for reducing N losses due to the waterlogging stress in poorly drained soils (Motavalli et al., 2003).

As with variable rate N application, multi-hybrid planting is becoming increasingly possible in many of the main agricultural areas of the United States. Multi-hybrid planting allows producers to place hybrids with traits that match particular areas of a field. For example, planting flood-tolerant hybrids in low landscape positions in South Dakota that are prone to waterlogging and planting drought-tolerant hybrids at the upper landscape positions resulted in an increase in yields (Sexton, Prairie, & Anderson, 2013; Sexton, Prairie, & Anderson, 2014). However, there is very limited information available on the use of multiple hybrid planting for agricultural fields prone to abiotic stresses such as flooding or drought. Further research is needed in the use of multiple hybrid planting in agricultural fields of variable soil characteristics and to evaluate the economic benefit of using this practice on farm scale. Delin and Berglund (2005) concluded that poor-performing areas in agricultural landscapes including areas that are susceptible to waterlogging during wet years can be better managed with precision agriculture techniques

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utilizing variable-rate and/or source N application in conjunction with variable hybrid and plant population plantings. Additionally, the decision making of the split application of N to the crop can be based on the remote sensing imagery obtained from unmanned aerial vehicles in season and targeted areas can be aerial sprayed with fertilizer mixes to enhance yields.

7.5 | Crop modeling and decision support systems

There are many models that can simulate crop growth and yield in response to soil waterlogging such as SWAGMAN Destiny (Meyer, Godwin, & White RJG, 1996), DRAINMOD (Skaggs, 2008), and the Agricultural Production Systems Simulator (APSIM; Asseng, Keating, Huth, & Eastham, 1997). These models can be used to assess the impact of changes in management practices on alleviation of waterlogging stress on crop plants and identifying areas or conditions that will cause yield reduction. For example, Bassu, Asseng, Motzo, and Giunta (2009) used APSIM-Wheat to predict waterlogging effect on wheat at multiple planting dates, and found that an early planting date could increase crop yield only in areas having low to moderate waterlogging risk, but did not increase crop yield in frequently waterlogged areas. However, adequacy of the process representations in these simulation models determines their success for use in estimating waterlogging stress (Shaw et al., 2013). In addition, GIS and remote sensing can be used for identifying areas in field that are vulnerable to soil waterlogging or drought conditions and can help in precision placement of any crop or nutrient management practice to reduce waterlogging stress. For example, using cover crops only in areas of maximum nutrient losses might result in reducing cost of cover crop planting and saving money for farmers. There is need for decision support tools for producers for decision making about the implementation of precision placement of different crop management practices for alleviation of soil waterlogging stress. McLellan et al. (2018) described Right Practice, Right Place (RPRP) Toolbox which includes a set of online conservation planning tools can link "right conservation practice" to "right place" at regional, watershed, and field scale, that will help in increasing the efficiency and effectiveness of water quality improvement efforts. Merriman, Daggupati, Srinivasan, and Hayhurst (2019) used Water Assessment Tool (SWAT) model to evaluate the effectiveness of use of multiple agricultural Best Management Practices (BMPs) either as single BMP or combination of BMPs on nutrient losses from an agricultural field. Although these decision support tools are evaluated for determining BMPs for water quality improvement, they have not yet tested for assessing effectiveness of BMPs for mitigating waterlogging stress in different soil and environmental conditions. Such models can act as tools for crop producers to make informed decisions about adoption of any crop management practice in areas vulnerable to waterlogging stress. However, there is limited knowledge on use of decision support tools for site-specific stationing of management practices discussed in this review article in response to soil waterlogging, and future research efforts should consider further the development of crop models along with decision support tools for soil waterlogging conditions.

8 | CONCLUSIONS

Soil waterlogging adversely affects plant growth and is responsible for major crop yield reductions globally. Crop losses to soil waterlogging are predicted to increase in many areas due to changing temperature and precipitation patterns associated with climate change. Plants generally have the ability to develop some adaptive traits, such as the growth of aerenchyma tissue or adventitious roots to withstand soil waterlogging stress. However, not all commercial cultivars are tolerant to waterlogging stress, and yield losses can be expected from such cultivars. In addition to affecting the growth and yield of major crops, soil waterlogging also results in N losses, reduced nutrient uptake and N use efficiency. There are limited studies that have evaluated possible management and recovery strategies to reduce losses from waterlogging stress, particularly for crops such as corn and soybean. Potential management strategies are summarized in this paper that can be considered by producers and land managers to improve yield and reduce N losses under waterlogged soil conditions. However, adoption of any management practices will be area-specific depending the ease of its use in current management plans followed by the producers. For example, raised beds could be used in area where producers utilize tillage for crop production. Raised beds for better drainage may not be adopted by no-till producers. More research is needed on possible adaptive options, such as N application to plants before, during, or after waterlogging, while also decreasing potential NO_3 runoff. A combination of management practices including managed subsurface drainage with N fertilizers and precision N applications with flood-tolerant varieties can also be employed to reduce soil waterlogging stress. Also, adoption of different management practices depends upon the waterlogging durations. Fields experiencing little waterlogging may only need to use tolerant hybrids, whereas fields experiencing heavy and multiple waterlogging events might have to consider using raised beds or tile drainage to avoid yield losses from the waterlogging. Under such conditions, crop modeling, along with decision support systems, should be developed for precision placement of BMPs to palliate waterlogging stress, as these crop models take into account the soil and environmental conditions

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prevalent in any particular field or watershed. These decision support systems will help producers in decision making on implementation of single or combination of BMPs on their agricultural fields suffering from waterlogging stress. Further research should be focused on cost-benefit analysis of these management practices for validating their economic viability and for developing future management plans that promote sustainable crop yields from waterlogged soils of variable duration and intermittency.

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